

ENVS-193DS Final

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[Link to repo](#)

```
# read in packages
library(tidyverse)
library(here)
library(DHARMA)
library(ggeffects)

# reading in data and creating new object
nest <- read_csv(
  here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

Problem 2. Data Analysis

a.

```
# creating clean data frame
nests_clean <- nest |>
  # selecting desired columns
  select(ant_species, height_cm, case_control) |>
  # filtering for desired ant species
  filter(ant_species %in% c("RRB", "BBR", "AB")) |>
  # creating new column for full scientific name
  mutate(sci_name = case_when(
    ant_species == "RRB" ~ "Crematogaster sjostedti",
    ant_species == "BBR" ~ "Crematogaster mimosae",
    ant_species == "AB" ~ "Crematogaster nigriceps"),
    # reordering factors for ant species
    ant_species = factor(ant_species,
      levels = c("RRB", "BBR", "AB"))
  )

# displaying structure of data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species : Factor w/ 3 levels "RRB","BBR","AB": 1 1 1 1 3 1 2 1 3 2 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ sci_name    : chr [1:270] "Crematogaster sjostedti" "Crematogaster sjostedti" "Crematogaster sjostedti" ...
```

```
# slicing sample
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
  ant_species height_cm case_control sci_name
  <fct>         <dbl>         <dbl> <chr>
1 BBR           318.             1 Crematogaster mimosae
2 RRB           228             1 Crematogaster sjostedti
```

3 AB	283	1 <i>Crematogaster nigriceps</i>
4 RRB	210	0 <i>Crematogaster sjostedti</i>
5 RRB	243	0 <i>Crematogaster sjostedti</i>
6 RRB	175	0 <i>Crematogaster sjostedti</i>
7 RRB	200.	0 <i>Crematogaster sjostedti</i>
8 AB	122	0 <i>Crematogaster nigriceps</i>
9 RRB	74	0 <i>Crematogaster sjostedti</i>
10 BBR	214	1 <i>Crematogaster mimosae</i>

b.

For this data, a value of 1 in the case_control column indicates that a nest was observed in the acacia tree in question. A value of zero in that said column means that no bird nest was observed in the tree.

c.

Tree Height (height_cm)

Tree height in this data set is a continuous variable that is measured in centimeters. As tree height increases the probability of nest occurrence would likely increase. Taller trees are more structurally complex and therefore have more nesting opportunities, they may also provide better predator avoidance for birds (Introduction, paragraph 2; Methods, “Study site and species” subsection, paragraph 1).

Acacia ant species (ant_species)

As ant species changes, the probability of nest occurrence would likely differ. This is due to each ant species exhibiting a distinct aggression level and mutualistic relationship with the host Acacia tree which affects the level of predator protection a nest in said tree would receive. (Introduction, paragraph 3; Methods, “Nest box experiment” subsection, paragraph 1)

d.

Model number	Ant Species	Tree Height	Predictor list
1	X	X	all predictors, no interaction (null model)
2	X	X	all predictors, with interaction (full model)

e.

f.

```
# running AIC test to determine best model
AIC(model1, model2)
```

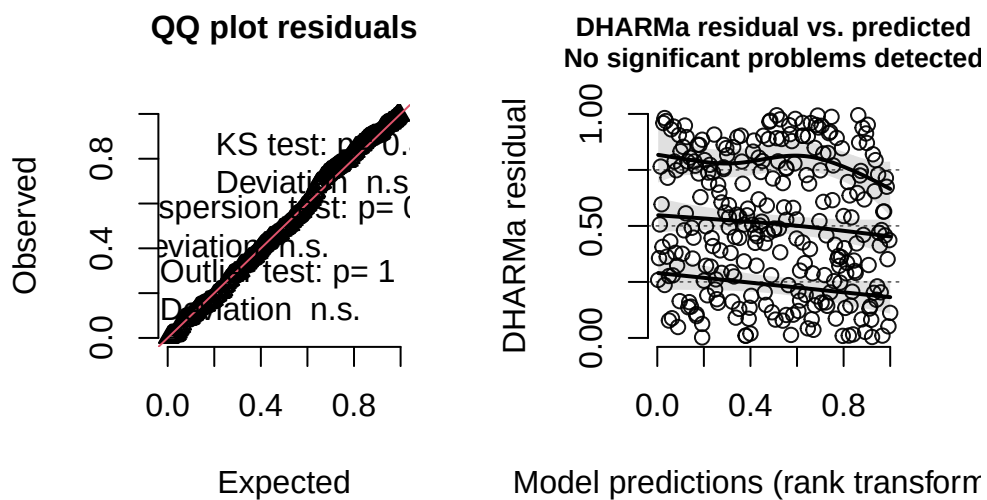
	df	AIC
model1	4	258.7430
model2	6	237.8042

The best model as determined by Akaike's Information Criterion (AIC) was the model that predicted test_case (whether or not a sample tree had a nest) using the interaction of tree height (height_cm) and species of acacia ant (ant_species)

g.

```
# plotting simulated residuals from DHARMA package.
plot(
  simulateResiduals(model2)
)
```

DHARMA residual



h.

```
# creating prediction model for model2
preds <- ggpredict(model2,
                    # terms for model, all tree heights all species
                    terms = c("height_cm[all]", "ant_species"))

# setting custom species colors
sp_cols <- c(
  "RRB" = "tan",
  "BBR" = "orange",
  "AB" = "lightblue3"
)

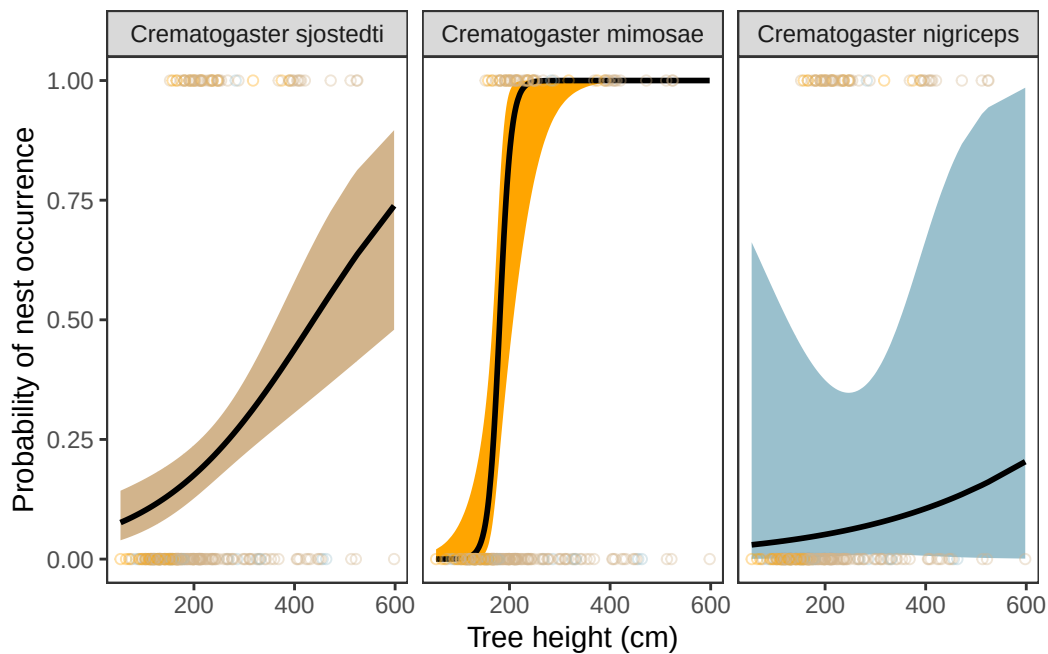
# creating facet labels from nest_clean
sp_labels <- nests_clean |>
  # choosing correct columns
  distinct(ant_species, sci_name) |>
  deframe()

# base layer: ggplot
ggplot() +
  # first layer: model confidence ribbon
  geom_ribbon(data = preds, # prediction data frame
            aes(x = x,
                ymin = conf.low, # min by low confidence
                ymax = conf.high, # max by high confidence
                fill = group)) + # filling color by group
  # second layer: model probability line
  geom_line(data = preds, # prediction data frame
            aes(x = x,
                y = predicted),
            # coloring line black
            color = "black",
            # making line thicker
            linewidth = 1) +
  # third layer: jitter for underlying data
  geom_jitter(data = nests_clean, # nests_clean data frame
             aes(x = height_cm, # x-axis: tree height
                 y = case_control, # y-axis: nest occurrence
                 color = ant_species), # coloring by species
             alpha = 0.35, # making points more opaque
```

```

    shape = 21, # changing shape
    height = 0) + # removing vertical jitter
# seperating by species
facet_wrap(~ group,
           # labelling with sci_name
           labeller = labeller(group = sp_labels)) +
# setting colors by species
scale_color_manual(values = sp_cols) +
scale_fill_manual(values = sp_cols) +
# relabelling x and y-axes
labs(
  x = "Tree height (cm)",
  y = "Probability of nest occurrence"
) +
# changing theme
theme_bw()+
# removing legend
theme(legend.position = "none",
      # removing gridlines
      panel.grid = element_blank())

```



i.